

DESIGN THINKING

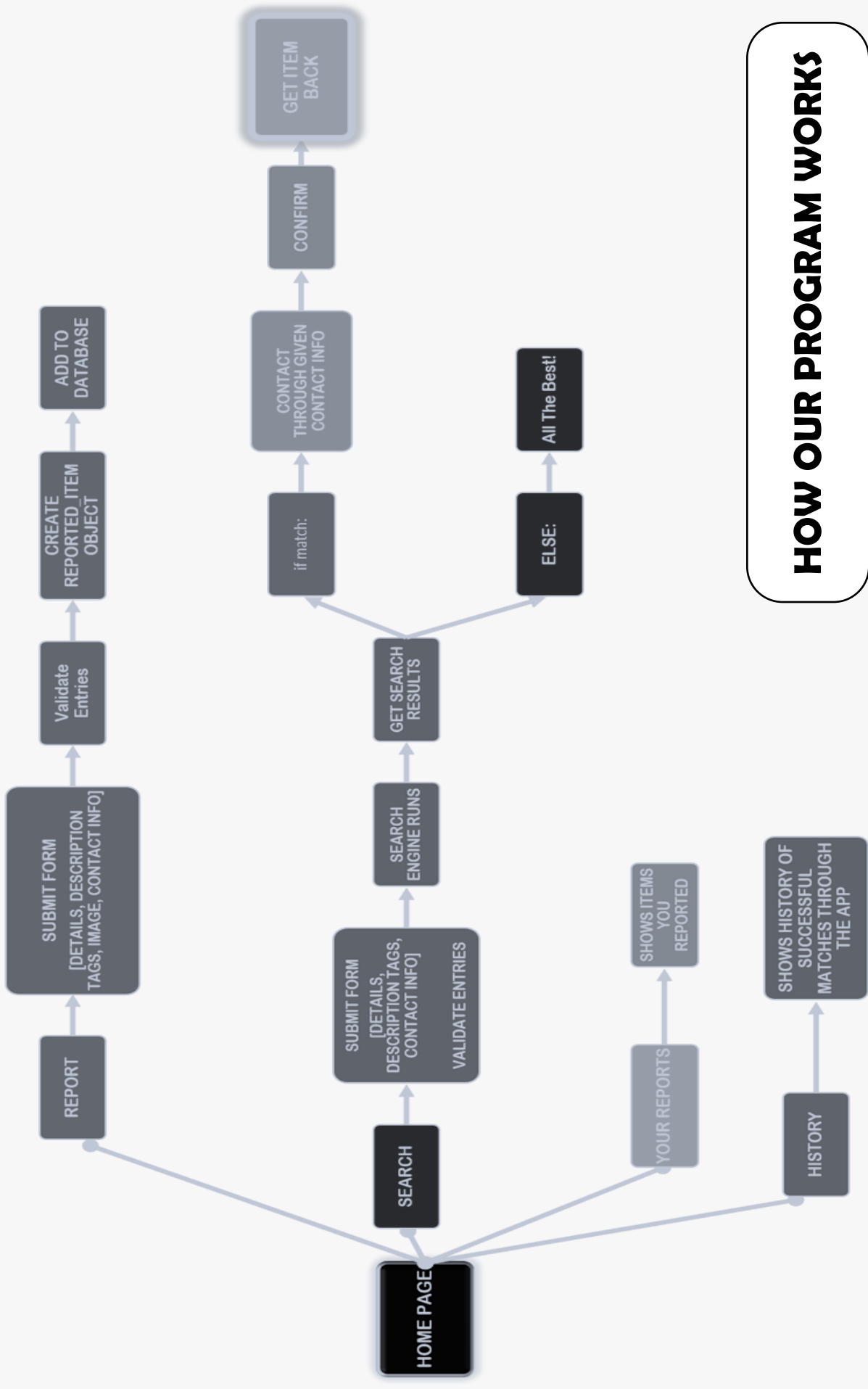
CAT-1 ASSIGNMENT

PROBLEM STATEMENT

- **Students often loose their belongings in the vast college campus.**
- **Finding it again takes quite the effort**
- **Brute force searching is not always a preferable option**
- **Therefore, there must exist a system where students can report items they find in random places, while the person who lost it can get to know about it through the same system, in a much more reliable manner**

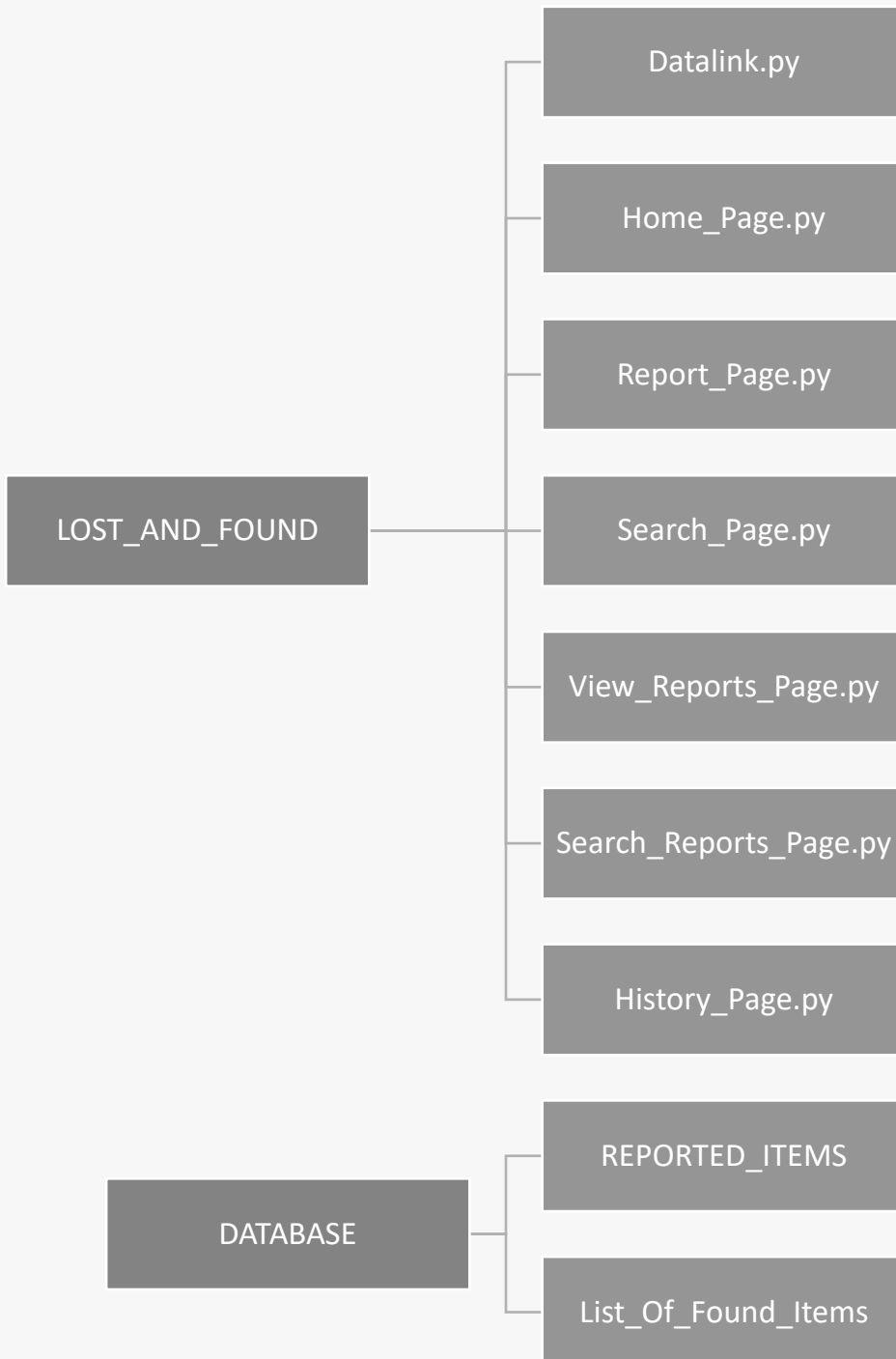
HOW OUR PROGRAM WORKS

- 1) Our program allows a passer-by to report a found item and provides the owner with a robust search engine to locate it.
- 2) The reporter can submit details of the found item, including the area, approximate date and time, descriptive tags, and a picture of the item.
- 3) The searcher can enter the approximate area, date, and time of losing the item. They must select relevant pre-existing tags from the available list and can also add their own custom tags.
- 4) The search engine analyzes the provided details and tags to identify potential matches and displays the matching reports in order of relevance.



HOW OUR PROGRAM WORKS

FILE STRUCTURE



Datalink.py

Home_Page.py

```
Cat-1 Assignment > Home_Page.py > get_old_reports
1 ***#//////////////////////////////////////[[<STARTS>]]\\
2
3 #\\//////////////////////////////////////[[<IMPORTS>]]\\
4
5 import tkinter as HUD
6 import sys
7 import os
8
9 import Datalink
10 import History_Page
11 import Report_Page
12 import Search_Page
13 import View_Reports_Page
14
15 #\\//////////////////////////////////////[[<FUNCTIONS>]]\\
16
17 def Quit():
18     Datalink.disconnect_datalink()
19     if os.path.exists("Temp"):
20         shutil.rmtree("Temp")
21     os.makedirs("Temp")
22     screen.destroy()
23     sys.exit()
24
25 #\\//////////////////////////////////////[[Func-End]]\\
26
27 def toggle_fullscreen(event=None):
28     current = screen.attributes('-fullscreen')
29     screen.attributes('-fullscreen', not current)
30
31 #\\//////////////////////////////////////[[Func-End]]\\
32
33 class Old_Reports:
34
35     def __init__(self, report_data):
36         self.id = report_data[0]
37         self.date = report_data[1]
38         self.time = report_data[2]
39         self.area = report_data[3]
40         self.tags = report_data[4].split(',')
41         self.email = report_data[5]
42         self.phone = report_data[6]
43         self.picture_path = report_data[7]
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48 def get_old_reports():
49     datalink_cursor = Datalink.get_cursor()
50     query = "SELECT * FROM Reported_Items"
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```

Report_Page.py

```
Cat-1 Assignment > Report_Page.py > ...
1  """
2  #//////////////////////////////////////////////////////////////////[[<START>]]//////////////////////////////////////////////////////////////////
3  #\\-----[[<IMPORTS>]]-----\\-----
4
5  import datalink
6  import tkinter as HUD
7  from tkinter import filedialog
8  import os
9  import shutil
10 import re
11 from datetime import date
12 import View_Report_Page
13
14 #\\-----[[<FUNCTIONS>]]-----\\-----
15
16 def report(home_page, report_page, view_reports_page, screen):
17
18     #\\-----[[<CLASS>]]-----\\-----
19
20     class Reported_Item():
21
22         def __init__(self, entries):
23
24             self.date = entries['date_entry']
25             self.time = entries['time_entry']
26             self.area = entries['area_entry']
27             self.tags = entries['desc_tags_entry']
28             self.email = entries['email_entry']
29             self.phone = entries['phone_entry']
30             self.picture_path = entries['picture_path']
31
32             date_sql = f"%d-%d-%d"%(self.date[1]:02d},{self.date[1]:02d},{self.date[0]:02d}"
33             time_sql = f"%d-%d-%d"%(self.time[1]:02d},{self.time[1]:02d},{self.time[0]:00}"
34
35             insert_query = """
36             INSERT INTO Reported_Items
37             (Date, Time, Area, Desc_Tags, Email, Phone, Picture_Path)
38             VALUES (%s, %s, %s, %s, %s, %s, %s)
39             """
40
41             insert_values = (
42                 date_sql,
43                 time_sql,
44                 self.area,
45                 ", ".join(self.tags),
46                 self.email if self.email else None,
47                 self.phone if self.phone else None,
48                 self.picture_path
49             )
50
```

```
Cat-1 Assignment > Report_Page.py > Report_Item > _init_
51     cursor = Datalink.get_cursor()
52     cursor.execute(insert_query, insert_values)
53
54     self.id = cursor.lastrowid
55
56     # ----- MOVE PICTURE -----
57
58     os.makedirs("Pictures", exist_ok=True)
59     old_name = os.path.basename(self.picture_path)
60     new_name = f"{self.id}_{old_name}"
61     new_path = os.path.join("Pictures", new_name)
62     shutil.move(self.picture_path, new_path)
63     self.picture_path = new_path
64
65     # ----- UPDATE PICTURE PATH -----
66
67     update_query = """
68     UPDATE Reported_Items
69     SET Picture_Path = %s
70     WHERE Id = %s
71     """
72
73     cursor.execute(update_query, (self.picture_path, self.id))
74     Datalink.commit()
75     Datalink.disconnect_datalink()
76     report_page.grid_remove()
77
78     @classmethod
79     def from_database(cls, report):
80         obj = cls._new_(cls)
81         obj.id = report[0]
82         obj.date = report[1]
83         obj.time = report[2]
84         obj.area = report[3]
85         obj.tags = report[4].split(", ")
86         obj.email = report[5]
87         obj.phone = report[6]
88         obj.picture_path = report[7]
89         return obj
90
91     """#\\-----[[<CLASS-END>]]-----\\-----"""
92
93     #setign up old reports
94     def get_old_reports():
95         datalink_cursor = Datalink.get_cursor()
96         query = "SELECT * FROM Reported_Items"
97         datalink_cursor.execute(query)
98         your_reports = datalink_cursor.fetchall()
99         your_report_objs = []
100
```


Report_Page.py

```
Cat-1 Assignment > Report_Page.py > ...
101 for report_data in your_reports:
102     your_report_objs.append(Reported_Item.from_database(report_data))
103
104     return your_report_objs
105
106
107 your_report_objs = get_old_reports()
108
109 #bring in the report page
110 home_page.grid_remove()
111
112 #Scaling for maximize
113 report_page.grid_configure(row=0, column=0, sticky='nsew')
114 report_page.grid_columnconfigure(0, weight=1)
115 report_page.grid_columnconfigure(1, weight=0)
116 report_page.grid_rowconfigure(2, weight=1)
117 report_page.grid_rowconfigure(10, weight=3)
118 report_page.configure(bg='black')
119
120 #Header
121 HUD.Label(report_page, text= "Report Found Item", font= ("Berlin Sans FB Demi", 25), bg='black', fg= 'orange'
122
123 #-----#]-----[FORM-Srt]-----[#
124
125 #-----#]-----[Defaults-Srt]-----[#
126
127 midspace = HUD.Label(report_page, text = ' ', font= ("OCR A Extended", 18), bg= 'black', fg= 'white', )
128
129 default_label_dict = {
130     'master': report_page,
131     'font': ("OCR A Extended", 12),
132     'bg': 'black',
133     'fg': 'white',
134     'anchor': 'w',
135     'width': 75
136 }
137
138 default_label_grid_dict = {
139     'column': 0,
140     'pady': (15,0),
141     'pady': 10
142 }
143
144 default_entry_dict = {
145     'master': report_page,
146     'font': ("OCR A Extended", 12),
147     'bg': 'black',
148     'fg': 'white',
149     'width': 75
150 }
151
152 default_entry_grid_dict = {
153     'column': 2,
154     'pady': (0,15)
155 }
156
157 #-----#]-----[Defaults-End]-----[#
158
159 #Date
160 HUD.Label(text="On which date did you find the item <dd/mm/yy> format: ", **default_label_dict).grid(row=1,
161 midspace.grid(row=1, column=1)
162 date_entry = HUD.Entry(**default_entry_dict)
163 date_entry.grid(row=1, **default_entry_grid_dict)
164
165 #Time
166 HUD.Label(text="At what time did you find the item <hh:mm> format: ", **default_label_dict).grid(row=2, **d
167 midspace.grid(row=2, column=1)
168 time_entry = HUD.Entry(**default_entry_dict)
169 time_entry.grid(row=2, **default_entry_grid_dict)
170
171 #Area
172 HUD.Label(text="In which area did you find the item: ", **default_label_dict).grid(row=3, **default_label_grid
173 midspace.grid(row=3, column=1)
174 area_entry = HUD.Entry(**default_entry_dict)
175 area_entry.grid(row=3, **default_entry_grid_dict)
176
177 #Desc Tags
178 HUD.Label(text="Describe item using relevant coma-separated tags: ", **default_label_dict).grid(row=4, **defa
179 midspace.grid(row=4, column=1)
180 desc_tags_entry = HUD.Entry(**default_entry_dict)
181 desc_tags_entry.grid(row=4, **default_entry_grid_dict)
182
183 #Contact
184 HUD.Label(text="Enter either [Email] or [Phone Number] or [Both]", **default_label_dict).grid(row=5, **default
185 midspace.grid(row=5, column=1)
186
187 #Email
188 HUD.Label(text="Enter your email address: ", **default_label_dict).grid(row=6, **default_label_grid_dict)
189 midspace.grid(row=6, column=1)
190 email_entry = HUD.Entry(**default_entry_dict)
191 email_entry.grid(row=6, **default_entry_grid_dict)
192
193 #Phone
194 HUD.Label(text="Enter your phone number: ", **default_label_dict).grid(row=7, **default_label_grid_dict)
195 midspace.grid(row=7, column=1)
196 phone_entry = HUD.Entry(**default_entry_dict)
197 phone_entry.grid(row=7, **default_entry_grid_dict)
198
199 #Picture
200 HUD.Label(text="Enter your picture path: ", **default_label_dict).grid(row=8, **default_label_grid_dict)
201 picture_path = HUD.StringVar()
```

Report_Page.py

```

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def choose_picture():
    selected_path = filedialog.askopenfilename(
        title="Select Item Picture",
        filetypes=[
            ('Image Files', '*.jpg *.jpeg *.png *.gif'),
            ('All Files', '*', '*')
        ])
    if selected_path:
        os.makedirs("Temp", exist_ok=True)
        file_name = os.path.basename(selected_path)
        destination = os.path.join("Temp", file_name)
        shutil.copy2(selected_path, destination)
        picture_path.set(destination)

HUD.Label(
    text="Picture of found item:",
    **default_label_dict
).grid(row=8, column=0, padx=(15, 0), pady=10)
midspace.grid(row=8, column=1)

HUD.Button(
    master=report_page,
    text="Choose Picture",
    command=choose_picture
).grid(row=8, column=2, padx=(0, 15))

HUD.Label(
    master=report_page,
    textvariable=picture_path,
    font=("OCR A Extended", 10),
    fg="black",
    bg="white",
    anchor="w"
).grid(row=9, column=2, padx=(0, 15), sticky="w")

#logging values and submit button
entries = dict()
entries['date_entry'] = date_entry
entries['time_entry'] = time_entry
entries['area_entry'] = area_entry
entries['desc_tags_entry'] = desc_tags_entry
entries['email_entry'] = email_entry
entries['phone_entry'] = phone_entry
entries['picture_path'] = picture_path

HUD.Button(
    master=report_page,
    text="Submit",
    command=lambda: submit(entries)
)

```

Report_Page.py

```

296 time_result = _isvalid_time(entries['time_entry'].get())
297
298 if time_result is None:
299     errors.append("Invalid time")
300 else:
301     valid_entries['time_entry'] = time_result
302
303 # ----- AREA -----
304
305 area = entries['area_entry'].get().strip()
306
307 if not area:
308     errors.append("Invalid area")
309 else:
310     valid_entries['area_entry'] = area
311
312 # ----- TAGS -----
313
314 tags = [
315     tag.strip()
316     for tag in entries['desc_tags_entry'].get().split(',')
317     if tag.strip()
318 ]
319
320 if not tags:
321     errors.append("Invalid description/tags")
322 else:
323     valid_entries['desc_tags_entry'] = tags
324
325 # ----- CONTACT -----
326
327 email = entries['email_entry'].get().strip()
328 phone = entries['phone_entry'].get().strip()
329
330 if not email and not phone:
331     errors.append("Enter either an email address or phone number")
332 else:
333     valid_entries['email_entry'] = email
334     valid_entries['phone_entry'] = phone
335
336 # ----- PICTURE -----
337
338 picture_path = entries['picture_path'].get()
339
340 if not picture_path:
341     errors.append("No picture selected")
342 else:
343     valid_entries['picture_path'] = picture_path
344
345

```

Search_Page.py

Search_Page.py

Cat-1 Assignment > Search_Page.py > ...	Cat-1 Assignment > Search_Page.py > ...
101 # [Func-End]	151 default_entry_dict = {
102	152 'master': search_page,
103	153 'font': ('OCR A Extended", 12),
104	154 'bg': 'black',
105	155 'fg': 'white',
106	156 'width': 75
107	157 }
108	158
109	159 default_entry_grid_dict = {'column': 2, 'padx': (0,15)}
110	160
111	161 # [Func-End]
112	162
113	163 #Date
114	164 HUD.Label(text="On which date did you find the item (<dd/mm/yy> format, optional): ", **default_label_dict).grid(row=
115	165
116	166 date_entry = HUD.Entry(**default_entry_dict)
117	167 date_entry.grid(row=1, **default_entry_grid_dict)
118	168
119	169 #Time
120	170 HUD.Label(text="At what time did you find the item (<hh:mm> format, optional): ", **default_label_dict).grid(row=2,
121	171
122	172
123	173 time_entry = HUD.Entry(**default_entry_dict)
124	174 time_entry.grid(row=2, **default_entry_grid_dict)
125	175
126	176 #Area
127	177 HUD.Label(text="In which area did you find the item (optional): ", **default_label_dict).grid(row=3, **default_label
128	178
129	179
130	180
131	181 #Contact
132	182 HUD.Label(text="Enter either [Email] or [Phone Number] or [Both]",**default_label_dict).grid(row=4, **default_label
133	183
134	184
135	185
136	186 #Email
137	187 HUD.Label(text="Enter your email address: ", **default_label_dict).grid(row=5, **default_label_grid_dict)
138	188
139	189
140	190
141	191 #Phone
142	192 HUD.Label(text="Enter your phone number: ", **default_label_dict).grid(row=6, **default_label_grid_dict)
143	193
144	194
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Search_Page.py

Search_Page.py

[illegible]

View_Reports_Page.py

```

1  """
2  # ===== [STARTS] =====
3  # ===== [EXPORTS] =====
4
5  import tkinter as HUD
6  import Datalink
7  from PIL import Image, ImageTk, ImageDraw, ImageFont
8  import os
9
10 # ===== [FUNCTIONS] =====
11
12 def go_home(current_page, home_page):
13     current_page.grid_remove()
14     home_page.grid(row=0, column=0)
15
16 # ===== [Func-End] =====
17
18 def is_matched(id):
19     datalink_cursor = Datalink.get_cursor()
20     get_query = """
21     SELECT is_matched
22     FROM Reported_Items
23     WHERE Id = %s
24     """
25     id_value = (id,)
26
27     datalink_cursor.execute(get_query, id_value)
28     match_result = datalink_cursor.fetchone()[0]
29
30     set_query = """
31     UPDATE Reported_Items
32     SET is_matched = %s
33     WHERE Id = %s;
34     """
35     values = (not match_result, id)
36     datalink_cursor.execute(set_query, values)
37
38     Datalink.commit()
39     Datalink.disconnect_datalink()
40     return not match_result
41
42 # ===== [Func-End] =====
43
44 def toggle_matched(button, id):
45     new_status = is_matched(id)
46     if new_status:
47         button.config(text="-- Matched")
48     else:
49         button.config(text="o Matched")
50
51 # ===== [Func-End] =====
52
53 def create_report_image(obj, width=450, height=250):
54     # =====
55     # LOAD IMAGE
56     # =====
57
58     try:
59         if os.path.exists(obj.picture_path):
60             image = Image.open(obj.picture_path).convert("RGBA")
61         else:
62             image = Image.new("RGBA", (width, height), "black")
63
64     except Exception:
65         image = Image.new("RGBA", (width, height), "black")
66
67 # FIT IMAGE TO ENTIRE CARD
68
69 image_ratio = image.width / image.height
70 card_ratio = width / height
71
72 if image_ratio > card_ratio:
73     new_height = height
74     new_width = int(new_height * image_ratio)
75
76 else:
77     new_width = width
78     new_height = int(new_width / image_ratio)
79
80 image = image.resize((new_width, new_height), Image.Resampling.LANCZOS)
81
82 # CENTER CROP
83
84 left = (new_width - width) // 2
85 top = (new_height - height) // 2
86 image = image.crop((left, top, left + width, top + height))
87
88 # SEMI-TRANSPARENT INFORMATION PANEL
89
90 panel_height = 105
91 overlay = Image.new("RGBA", image.size, (0, 0, 0, 0))
92 overlay_draw = ImageDraw.Draw(overlay)
93 overlay_draw.rectangle(
94     (0, height - panel_height, width, height),
95     fill=(0, 0, 0, 150)
96 )
97
98 image = Image.alpha_composite(image.convert("RGBA"), overlay)
99 draw = ImageDraw.Draw(image)
100
101 # REPORT INFORMATION
102
103 report_text = (
104     f"ID: {obj.id}\n"
105     f"Date: {obj.date}\n"
106     f"Time: {obj.time}\n"
107     f"Area: {obj.area}\n"
108     f"Tags: {obj.tags}"
109 )
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View_Reports_Page.py

```

121 # =====
122 # FONT
123 # =====
124
125 try:
126     font = ImageFont.truetype("C:/Windows/Fonts/OCRABXT.TTF", 12)
127 except Exception:
128     font = ImageFont.load_default()
129
130 # BURN TEXT ONTO IMAGE
131 # =====
132 # =====
133
134 draw.multiline_text(
135     18, height - panel_height + 12,
136     report_text,
137     font=font,
138     fill="white",
139     spacing=4
140 )
141 return image
142
143 #]:=====:[Func-End]:=====
144
145 def view_reports(home_page, view_reports_page, screen, your_report_objs):
146
147     home_page.grid_remove()
148
149     # CLEAR OLD CONTENT
150     # =====
151     # =====
152
153     for widget in view_reports_page.winfo_children():
154         widget.destroy()
155     view_reports_page.grid_configure(row=0, column=0, sticky="nsew")
156     view_reports_page.configure(bg='black')
157
158     # =====
159     # TITLE
160     # =====
161
162     HUD_Label(
163         view_reports_page,
164         text="View Your Reports",
165         font=(Berlin Sans FB Demi , 25),
166         bg='black',
167         fg='cyan',
168         anchor="n",
169     ).grid(row=0, column=0, columnspan=2, padx=20, pady=(20, 25))
170
171     # =====
172     # SCROLLING AREA
173     # =====
174
175     canvas = HUD.Canvas(view_reports_page, bg='black', highlightthickness=0)
176     scrollbar = HUD.Scrollbar(view_reports_page, orient='vertical', command=canvas.yview)
177     canvas.configure(yscrollcommand=scrollbar.set)
178     canvas.grid(row=1, column=0, sticky='nsew')
179     scrollbar.grid(row=1, column=1, sticky='ns')
180
181     view_reports_page.grid_rowconfigure(1, weight=1)
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```

View_Reports_Page.py

```

Cat-1 Assignment > View_Reports_Page.py > is_matched
243
244
245 photo = ImageTk.PhotoImage(report_image)
246
247 # -----
248 # SINGLE IMAGE LABEL
249 # -----
250 picture_label = HUD.Label(card, image=photo, bg='black', bd=0)
251 picture_label.image = photo
252 picture_label.place(x=0, y=0, width=450, height=250)
253
254 # -----
255 # MATCHED BUTTON
256 # -----
257 matched_button = HUD.Button(
258     card,
259     text="\u25a1 Matched",
260     bg='grey',
261     fg='white',
262     font=("OCR A Extended", 8)
263 )
264
265 matched_button.config(
266     command=lambda button=matched_button, Id=obj.id:
267         | toggle_matched(button,Id)
268 )
269
270 # -----
271 # PLACE BUTTON AT SOUTH-EAST
272 # -----
273 matched_button.place(relx=0.97, rely=0.94, anchor='se')
274
275 # -----
276 # BACK BUTTON
277 # -----
278 HUD.Button(
279     view_reports_page,
280     text="Back",
281     bg='grey',
282     fg='white',
283     command=lambda: go_home(view_reports_page, home_page)
284 ).grid(row=3, column=0, columnsan=2, pady=15)
285
286 #].....[Func_End].....[#
287
288 """#]]<<-END->>[[

```

Search_Result_Page.py

Search_Result_Page.py

Search_Result_Page.py

```
151 Cat1 Assignment > Search_Result_Page.py > ...
152 canvas = HUD.Canvas(search_result_page, bg='black', highlightthickness=0)
153 scrollbar = HUD.Scrollbar(search_result_page, orient='vertical', command=canvas.yview)
154
155 canvas.configure(yscrollcommand=scrollbar.set)
156
157 canvas.grid(row=1, column=0, sticky='nsew')
158 scrollbar.grid(row=1, column=1, sticky='ns')
159
160 search_result_page.grid_rowconfigure(1, weight=1)
161 search_result_page.grid_columnconfigure(0, weight=1)
162
163 scrollbar_frame = HUD.Frame(canvas, bg='black')
164 scrollbar_frame = canvas.create_window(
165     (0, 0),
166     window=scrollbar_frame,
167     anchor='nw'
168 )
169
170 #Single-column layout
171 scrollbar_frame.grid_columnconfigure(0, weight=1)
172
173 def update_scroll_region(event):
174     canvas.configure(scrollregion=canvas.bbox("all"))
175
176 scrollbar_frame.bind("<Configure>", update_scroll_region)
177
178 def resize_scrollable_frame(event):
179     canvas.itemconfig(canvas_window, width=event.width)
180
181 canvas.bind("<Configure>", resize_scrollable_frame)
182
183 for no, obj in enumerate(search_item.search_result):
184     row = no
185     column = 0
186
187     card = HUD.Frame(
188         scrollbar_frame,
189         bg='black',
190         bd=4,
191         relief='raised',
192         width=450,
193         height=250
194     )
195
196 card.grid(
197     row=row,
198     column=column,
199     padx=15,
200     pady=15,
```

```
201 Cat1 Assignment > Search_Result_Page.py > Search_result
202 sticky='n'
203 )
204
205 card.grid_propagate(False)
206
207 report_image = create_report_image(
208     obj,
209     width=450,
210     height=250
211 )
212
213 photo = ImageTk.PhotoImage(report_image)
214
215 picture_label = HUD.Label(
216     card,
217     image=photo,
218     bg='black',
219     bd=0
220 )
221
222 picture_label.image = photo
223
224 picture_label.place(
225     x=0,
226     y=0,
227     width=450,
228     height=250
229 )
230
231 #Match percentage
232 match_percent = int(round(obj.match_percent))
233
234 match_label = HUD.Label(
235     card,
236     text=f"{match_percent}% MATCH",
237     bg='black',
238     fg='white',
239     font=("OCR A Extended", 10)
240 )
241
242 match_label.place(
243     x=12,
244     y=257
245 )
246
247 #Match health bar
248 bar_x = 12
249 bar_y = 278
250 bar_width = 300
251 bar_height = 8
```

```
252 Cat1 Assignment > Search_Result_Page.py > Search_result
253 text=f"{match_percent}% MATCH",
254 bg='black',
255 fg='white',
256 font=("OCR A Extended", 10)
257 )
258
259 match_label.place(
260     x=12,
261     y=257
262 )
263
264 #Match health bar
265 bar_x = 12
266 bar_y = 278
267 bar_width = 300
268 bar_height = 8
269
270 green_width = int(bar_width * match_percent / 100)
271 red_width = bar_width - green_width
272
273 HUD.Frame(
274     card,
275     bg='green',
276     width=green_width,
277     height=bar_height
278 ).place(
279     x=bar_x,
280     y=bar_y
281 )
282
283 if red_width > 0:
284     HUD.Frame(
285         card,
286         bg='red',
287         width=red_width,
288         height=bar_height
289     ).place(
290         x=bar_x + green_width,
291         y=bar_y
292     )
293
294 found_button = HUD.Button(
295     card,
296     text="Found",
297     bg='grey',
298     fg='white',
299     font=("OCR A Extended", 8)
300 )
301
302 found_button.config(
```

Search_Result_Page.py

```

283         found_button.config(
284             command=lambda button=found_button, Id=obj.id:
285                 found_item(button, Id)
286         )
287
288         found_button.place(
289             relx=0.97,
290             rely=0.94,
291             anchor='se'
292         )
293
294     HUD.Button(
295         search_result_page,
296         text="Back",
297         bg='grey',
298         fg='white',
299         command=lambda: go_home(search_result_page, home_page)
300     ).grid(
301         row=3,
302         column=0,
303         columnspan=2,
304         pady=15
305     )
306
307 """# //////////////////////////////////////

```


History_Page.py

FUTURE ENHANCEMENTS

- 1) 1. The application can be developed into a web or mobile application for easier access across different devices.
- 2) 2. An advanced image-matching system can be implemented to compare uploaded pictures and improve the accuracy of search results.
- 3) 3. Notifications can be added to alert users when a potential match for their lost or found item is identified.
- 4) 4. Location-based searching can be introduced using maps and GPS to improve the accuracy of area-based matching.
- 5) 5. User authentication can be added to provide secure accounts and protect user information.
- 6) 6. The search algorithm can be further enhanced using machine learning to provide more accurate and intelligent matching.



THANK YOU